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COMPUTER SECURITY WK5

1. scenario analysis

A software company releases a new application for users to download from its official website. To ensure that the software has not been changed or corrupted, the company uses hashing and digital signatures. First, a cryptographic hash function such as SHA-256 is used to create a unique hash value from the software file. Users can calculate the hash of their downloaded file and compare it with the hash published by the company. If the values match, the file is likely unchanged. The company can also use a digital signature to provide stronger verification. The software developer signs the file using a private key, while users verify the signature using the developer's public key. This confirms that the software came from the claimed developer and has not been modified. Hashing mainly checks data integrity, while digital signatures provide integrity, authentication, and non-repudiation. This process helps protect users from downloading modified or malicious software.

2.CONCEPT RESEARCH:

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hashing algorithm used to protect and verify digital information. It takes data of any size and produces a fixed-length 256-bit hash value, usually displayed as a 64-character hexadecimal string. Even a small change in the original data produces a completely different hash. SHA-256 is designed to be one-way, meaning it is extremely difficult to recover the original data from its hash. It is commonly used for checking file integrity, verifying software downloads, storing security-related information, and supporting digital signatures. Its main role is to provide reliable data integrity and detect unauthorized changes.

3.TOOL PRACTICE:

```
import hashlib
```

```
message = "I am a Computer Security student."
```

```
# Create SHA-256 hash
```

```
hash_value = hashlib.sha256(message.encode()).hexdigest()
```

```
print("Message:", message)
```

```
print("SHA-256 Hash:", hash_value)
```

In this practical activity, I used Python to create a SHA-256 hash of a message. The message was converted into bytes and processed using Python's hashlib library. The resulting hash was a fixed 64-character hexadecimal value. I observed that changing even one character in the message produced a completely different hash. This showed me how hashing can be used to verify the integrity of information. Although SHA-256 hashing alone is not a digital signature, it is an important part of many signature systems because the hash can be signed using a private key. This practical exercise helped me understand how cryptographic techniques protect digital information from unauthorized changes.

4.DIAGRAM:

